

SERGII KHOMENKO

Software Professional | Quality Engineering and Development

Montréal area, Quebec, Canada | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software engineer and quality specialist with 25+ years in technology, including 19+ years in software QA and automation. In recent years, I have spent most of my time building test platforms, backend services, CI tools, and full-stack products. I am comfortable taking a problem from design to deployment and putting the tests in place as part of that work.

CORE SKILLS

Backend and languages: Go, Python, TypeScript, JavaScript, SQL. REST, GraphQL, PostgreSQL, background jobs, migrations, authentication, and command-line tools.

Frontend: React, Next.js, Vite, React Router, FullCalendar, i18next.

CI, infrastructure, and cloud: Docker, Docker Compose, GitHub Actions, CircleCI, Jenkins, AWS, Linux, and systemd.

Testing: Playwright, PyTest, Go test, Vitest, and Postman. API, UI, integration, end-to-end, and performance coverage.

Applied AI: Ollama, RAG, pgvector, multilingual embeddings, and AI coding tools.

Daily tools: Git, GitHub, Jira, Confluence, PyCharm, and VS Code. Uses ADRs to document important technical decisions.

SELECTED PROJECTS

Booking and Scheduling Platform | Personal project

- Solo build, end-to-end: domain model, Go REST backend, React and TypeScript frontend, PostgreSQL, test suite, CI, and production deployment.
- Double bookings are blocked at three independent levels, so a failure in one layer cannot corrupt a customer's calendar.
- A background worker sends reminders exactly once, survives restarts, and re-arms itself when a booking moves.
- Handles real customer data: Google sign-in, expiring public booking links, single-use invitations, and a consent record that can be audited.
- Fully bilingual, French and English, across the interface and outgoing mail.
- Unit, API, and browser tests run on every pull request, and the full suite runs locally before any push.

Go, React, TypeScript, PostgreSQL, Docker, GitHub Actions, Playwright, pytest

Bilingual Document Assistant | Working demonstrator

- An AI assistant that answers questions about a long French policy document and cites the exact article it used, so the answer can be verified instead of trusted.
- Questions work in French or English. An English question still finds the right French passages, and the answer comes back in the language it was asked in.
- Runs entirely on local hardware. No part of the document leaves the machine, which is what makes it usable for internal material.
- Built a diagnostic view showing what the system retrieved and why, which tells you whether a wrong answer came from search or from the model.

Next.js, TypeScript, PostgreSQL with pgvector, Python, local LLMs

PROFESSIONAL EXPERIENCE

LIGHTSPEED COMMERCE | Montréal, Canada

Staff Quality Test Developer | Dec 2019 to 2026

- Worked on front-end, backend, API, and integration test frameworks for several point-of-sale and commerce products. Teams reused the same libraries and test infrastructure across projects.
- Connected the suites to CI/CD, added quality gates, and investigated unstable or failed runs. Also built REST and GraphQL test tooling and a reporting service that made CI results available through APIs.
- Planned coverage with developers, product managers, support, and QA. The work included estimates, acceptance criteria, risk reviews, and code reviews. Mentored more than five test developers on framework design and day-to-day engineering practices.
- Used AI coding assistants and local LLMs for implementation, code analysis, failure investigation, and documentation.

Selected Lightspeed achievements

- Built a failure-triage bot that used an LLM to sort failures into regressions, outdated tests, or flaky tests. It posted the result and suggested next steps on the pull request.
- Added a CI check that flagged code changes affecting functionality without matching automated tests.
- Set up AWS-based test infrastructure and built automation for POS, loyalty, fulfillment, analytics, and cloud backend services.
- Brought flaky test failures close to zero by monitoring runs, finding recurring causes, and hardening the framework.
- Cut CircleCI usage from about 18M to 3M credits per month. That was an 80% reduction, worth roughly USD100,000 per year at standard pricing.

X20 MEDIA | Montréal, Canada

QA Technical Lead | Oct 2015 to Dec 2019

- Managed a five-person QA team covering web, mobile, desktop, and digital-signage products. Coached the team and aligned the manual and automated testing work.
- Introduced API testing, designed the automation framework, and built the regression suite.
- Put the automated suites into CI and used metrics to judge release readiness and risk. Wrote test strategies and worked closely with development, research, product management, and support.

BLACKBERRY | Montréal, Canada

Software Test Specialist | Apr 2011 to Sep 2015

- Owned regression quality for calendar and personal information management features, covering release validation, defect triage, risk assessment, and performance testing.
- Automated recurring test work, expanded calendar coverage, and worked with product managers and developers on specifications and issue investigation. Several improvements I proposed were added to the product.

EARLIER EXPERIENCE

Tungle Inc., Software Developer, Test Group | Jan 2011 to Apr 2011

GlobalLogic, QA Lead | Feb 2007 to Oct 2010

Hoffmann-La Roche, IT Supporter | 2002 to 2007

Unitel, System Engineer and Software Developer | 2000 to 2002

EDUCATION

Bachelor's Degree in Engineering

Kyiv International University of Civil Aviation, Kyiv, Ukraine